

Lab Assignments 1

For practicing – no grading

ECE 253 - Fall 2025
Digital Image Processing
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Overview

This lab assignment covers the implementation of linear-phase FIR filters, FIR filter design using the window method, and the computation of the frequency response using the DFT. You will write code to design and analyze FIR filters, demonstrate linear-phase behavior, and apply filters to signals.

Task 1: Frequency Response using the DFT

Objective: Implement a function to compute the frequency response of an FIR filter using the DFT.

Steps:

1. Implementation:

- Write a function `compute_dft_response(filter_coefficients)` that:
 - Takes as input the FIR filter coefficients.
 - Uses the **DFT** to compute the frequency response of the filter (use `numpy.fft.fft`).
 - Returns the magnitude and phase response.

2. Apply your function to the FIR filter $H(z) = 1 + z^{-1}$.

3. Analysis:

- Plot the DFT-based magnitude and phase response.

Task 2: FIR Filter Design using the Window Method

Objective: Implement a low-pass FIR filter using the window method and analyze its linear-phase behavior.

Steps:

1. Design an FIR filter using the **Hamming window**:

- Choose a filter length $N = 51$.
- Set the desired cutoff frequency $\omega_c = 0.4\pi$ (normalized frequency).
- Use the Hamming window to compute the FIR filter coefficients.

2. **Implementation:**

- Write a function `fir_filter_window(N, omega_c)` that:
 - Takes the filter length N and the cutoff frequency ω_c as inputs.
 - Designs the low-pass FIR filter using the window method.
 - Returns the filter coefficients.

3. **Analysis:**

- Plot the **magnitude** and **phase responses** of the designed filter using the function designed in Task 1.
- Show that the filter has **linear phase** by plotting the phase response and confirming it is a straight line.

Task 3: Linear-Phase Behavior of FIR Filters

Objective: Explore and demonstrate the linear-phase property of FIR filters with symmetric coefficients.

Steps:

1. Design an FIR filter using the window method with **symmetric coefficients**:

- Choose $N = 41$ and cutoff frequency $\omega_c = 0.5\pi$.
- Ensure that the filter is designed with symmetric coefficients.

2. **Implementation:**

- Write a function `is_linear_phase(filter_coefficients)` that:
 - Takes as input the FIR filter coefficients.
 - Checks if the filter has linear phase by verifying symmetry in the filter coefficients.

- Returns **True** if the filter has linear phase, and **False** otherwise.

3. Analysis:

- Demonstrate the function's result by checking the linear-phase property of the designed filter.

Task 4: FIR Filter Application on a Signal using Convolution

Objective: Apply the designed FIR filter to a signal using time-domain convolution.

Steps:

1. Create a test signal:

- Generate a test signal composed of a sum of sinusoids at different frequencies (e.g., $x(n) = \sin(0.2\pi n) + \sin(0.7\pi n)$).

2. Implementation:

- Write a function `apply_fir_filter(signal, filter_coefficients)` that:
 - Convolves the input signal with the FIR filter using **time-domain convolution**.
 - Returns the filtered signal.

3. Analysis:

- Apply your FIR low-pass filter from **Task 2** to the test signal and plot both the original and filtered signals in the time domain.
- Discuss the impact of the filter on the high-frequency components of the test signal.